Econometric Analysis of Profitability, Liquidity, and Non-Debt Tax Shield on Capital Structure with Firm Size Moderator in Indonesian Property Sector

📌 Project Overview

This repository contains an econometric analysis conducted in Python that examines the firm size as the moderator factor in determining capital structure. Previous studies have produced mixed results regarding this relationship, motivating a closer investigation. Originally inspired by contemporary research, we explore the causal relationship between *profitability*, *liquidity*, and *non-debt tax shield* on *capital structure* while addressing *firm size* as the moderation feature. This project applies econometric techniques with evaluation key statistical assumptions, such as heteroskedasticity, auto-correlation, and multicollinearity.

Problem Statement

With all various theories in the financial domain, the determinants of capital structure in the Indonesian property sector remain empirically inconsistent. While profitability and liquidity are traditional drivers, their impact often fluctuates based on the scale of the firm. There is a lack of consensus on whether firm size acts as a catalyst or a buffer in these relationships. This study addresses this gap by investigating if firm size significantly moderates the effect of profitability, liquidity, and non-debt tax shields on the capital structure of companies under special monitoring.

Objectives

- **Model Explanability** to quantify the proportion of variance in capital structure explained by the financial predictors using the Coefficient of Determination (R^2)
- **Evaluate** individual (partial) and simultaneous (joint) significance of regression coefficients
- **Conduct** diagnostic test for multicollinearity, heteroskedasticity, and autocorrelation
- **Determine** the efficient panel estimator between Fixed and Random Effects models

Project Constraints

- The dataset was collected from the Indonesian Stock Exchange official site in a form of census.
- Data Scope:

    - 17 real estate companies listed under special monitoring,
    - 5 years observations period (2019-2023),
    - No missing value, resulting in 85 data entries.
- Statistical approach is descriptive with no concern on statistical inference, as the number of samples equals the population size.

📚 Literature Review & Theoretical Framework

This analysis is grounded on the related literatures as follows:

Title	Method	Description
The Effect of Profitability, Liquidity, and Non-Debt Tax Shield on Capital Structure with Company Size as a Moderator (2025).	Basic regression, hypothesis test, significance evaluation	The basis of exploration that expands the development analysis into econometrics
Introductory Econometrics: A Modern Approach	Basic regression, Pooled ordinary least square, Panel data methods	The primary reference in determining the appropriate regression model for the case and designing the model's structural formula
The Evaluation of Liquidity and Capital Structure of Manufacturing Firms in Nigeria	Pooled ordinary least square, hypothesis test, significance evaluation	The methodological reference in a proper statistical approach

As key literatures and frameworks are introduced above, we developed this econometrics analysis extending such:

1. Applying an econometric approach on a friend's previous thesis (first literature) that worked with basic regression techniques.
2. Conducting diagnostic analyses to examine the underlying statistical properties of the observations.

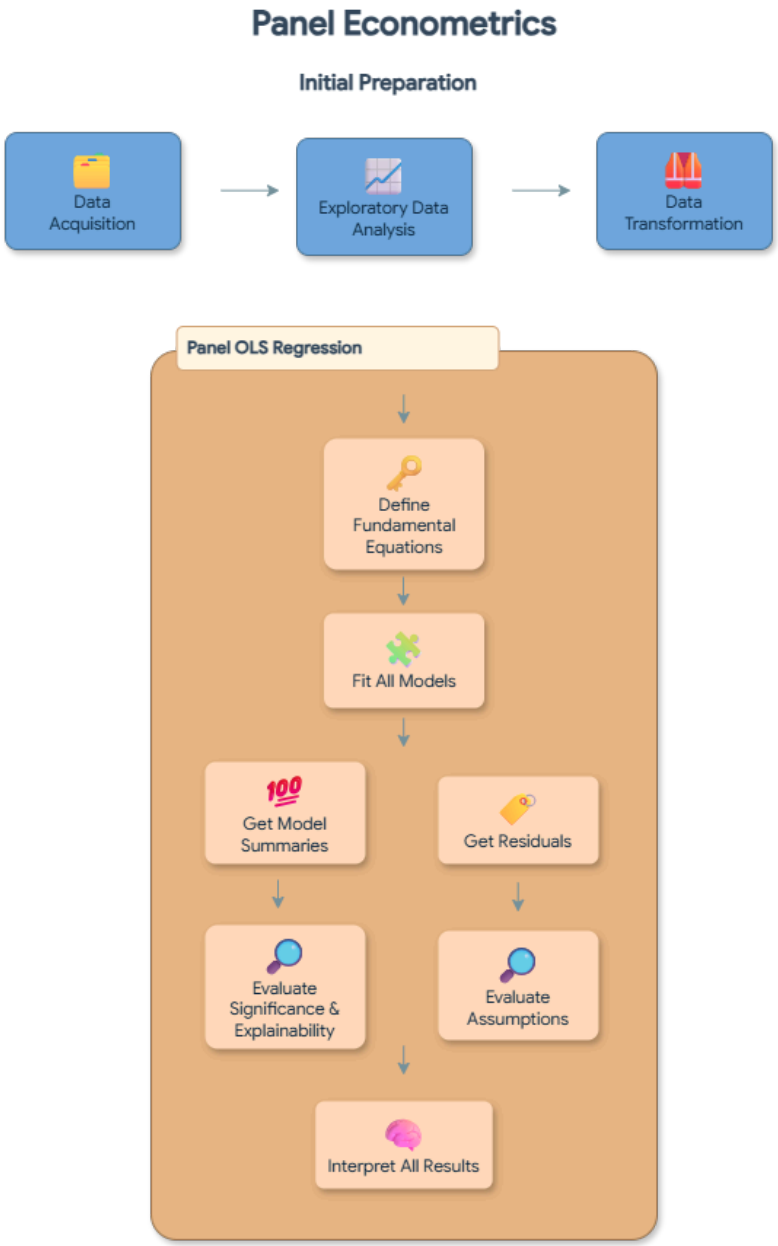
Variable Definition and Grounding

Below are described the financial variables introduced in this analysis.

Financial Variables	Description	Symbol	Formulas
Debt-to-Equity Ratio (proxy for capital structure)	Level of debt a company takes on to finance its operations, against the level of capital, or equity, that's available	DER	$\frac{Total\ Liabilities}{Total\ Equities}$
Net Profit Margin (proxy for profitability)	Metric representing the percentage of revenue remaining as profit after deducting all operating expenses, interest, taxes, and cost of goods sold (COGS)	NPM	$\frac{Net\ Profit}{Total\ Revenue}$
Current Ratio (proxy for liquidity)	Liquidity metric that measures a company's ability to pay short-term obligations (due within one year) with its short-term assets.	CR	$\frac{Current\ Assets}{Current\ Liabilities}$
Non-Debt Tax Shield	Reduction in taxable income for a corporation through claiming allowable deductions such as mortgage interest, medical expenses, charitable donations, amortization, and depreciation.	NDTS	$\frac{Depreciation}{Total\ Assets}$
Firm Size	Quantifiable measure of a business's operational scale, typically defined by employee headcount, annual revenue, or total assets	FS	$ln\ (Total\ Assets)$

🛠️ Methodology

The analytical workflow follows a rigorous econometric pipeline to ensure consistent estimates.



1. **Data Acquisition:** Sourced via Indonesian Stock Exchange website and calculate formulas
2. **Exploratory Data Analysis:** Performed visual analysis on data attributes
3. **Data Transformation:** Applied mean centering and Z-score standardization
4. **Panel OLS Regression:** Examined and evaluated the whole regression pipeline

Model Specification

The primary relationship is modeled as follows:

Basic Regression:

$$Y_{it} = \beta_0 + \beta_1 X_{1it} + \beta_2 X_{2it} + \beta_3 X_{3it} + \mu_{it}$$

Moderated Regression:

$$Y_{it} = \beta_0 + \beta_1 X_{1it} + \beta_2 X_{2it} + \beta_3 X_{3it} + \beta_4 (X_{1it} M_{it}) + \beta_5 (X_{2it} M_{it}) + \beta_6 (X_{3it} M_{it}) + \mu_{it}$$

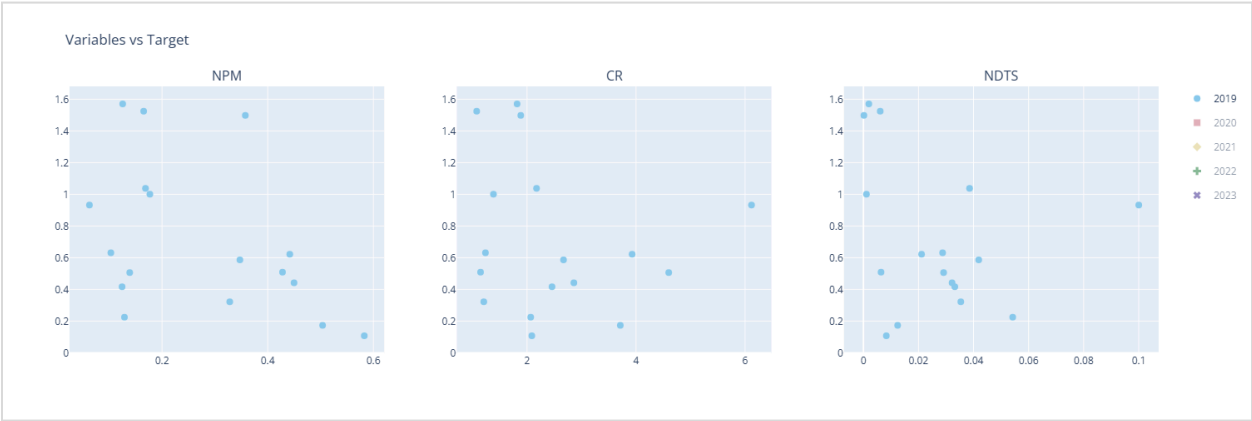
Where:

- X_1 : net profit margin (NPM)
- X_2 : current ratio (CR)
- X_3 : non-debt tax shield (NDTS)
- Y : debt-to-equity ratio (DER)
- M : firm size (FM)
- μ : error term
- i : firm (company)
- t : time period

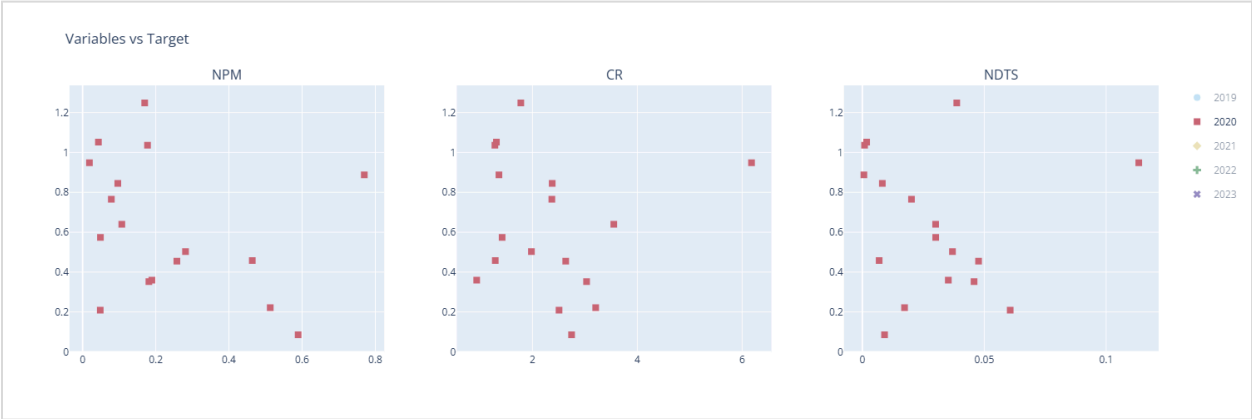
📊 Exploratory Data Analysis (EDA)

Comprehensive visualization of the dataset reveals key underlying patterns:

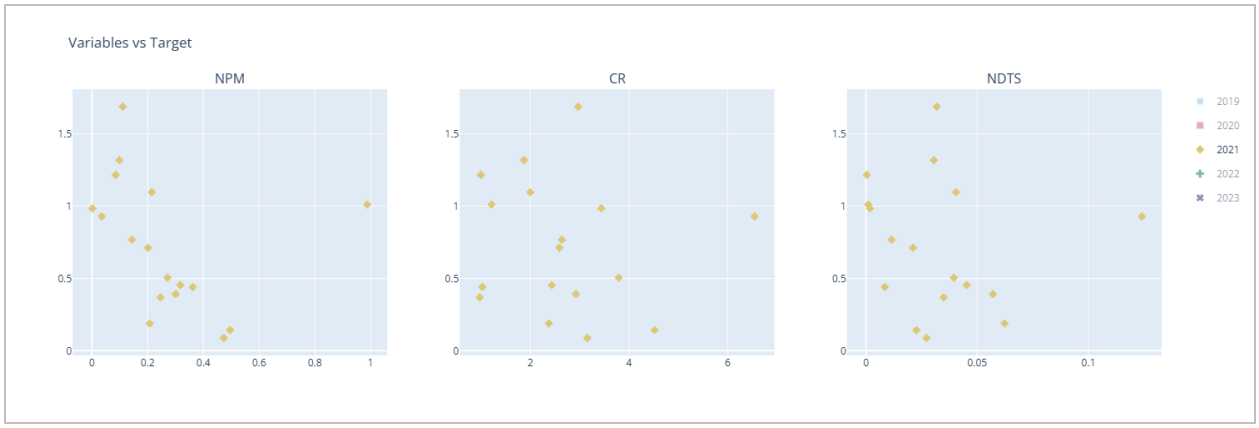
- **Independent-to-Target Variables Relationship:**



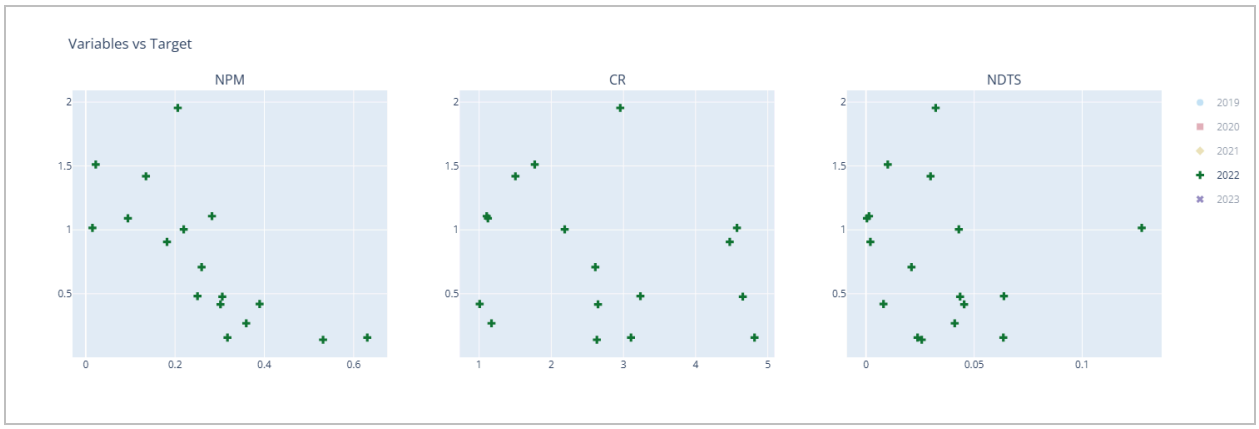
2019 period



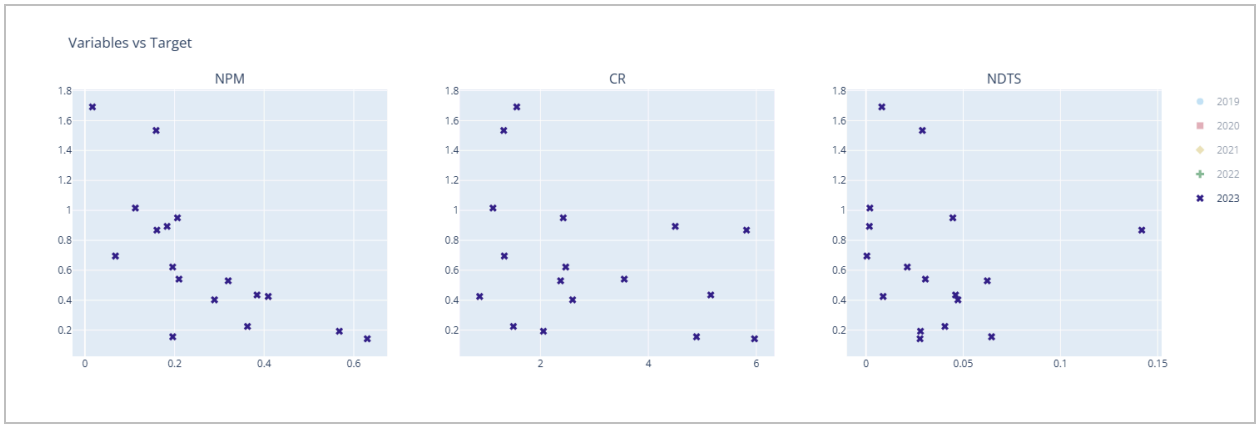
2020 period



2021 period

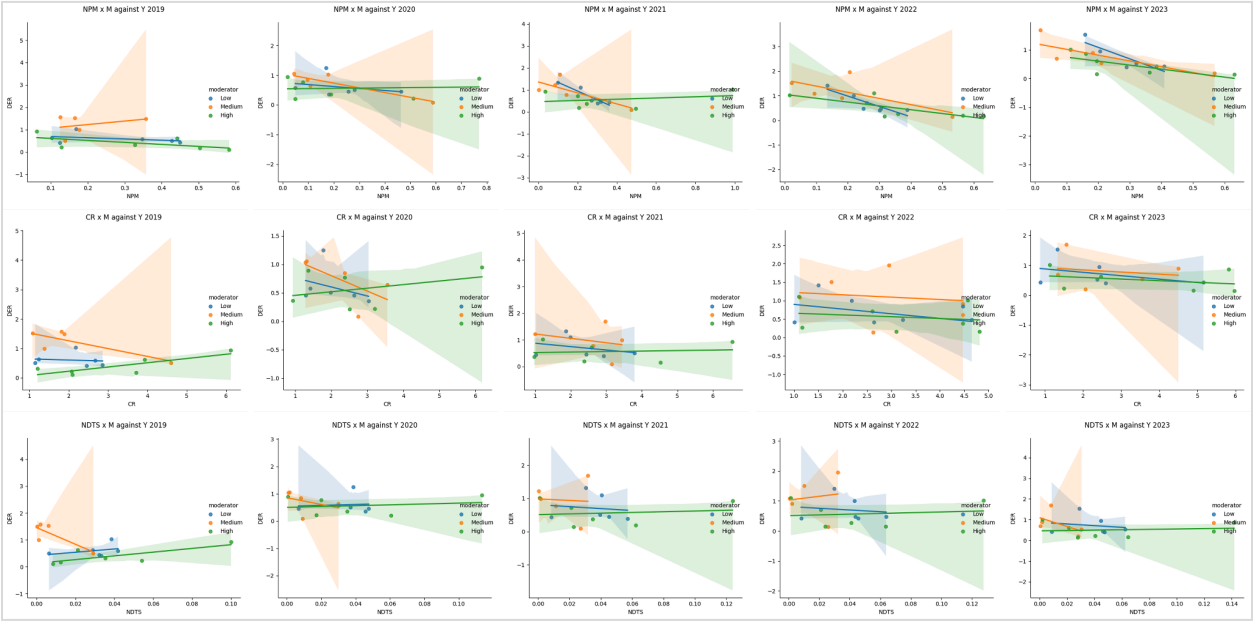


2022 period

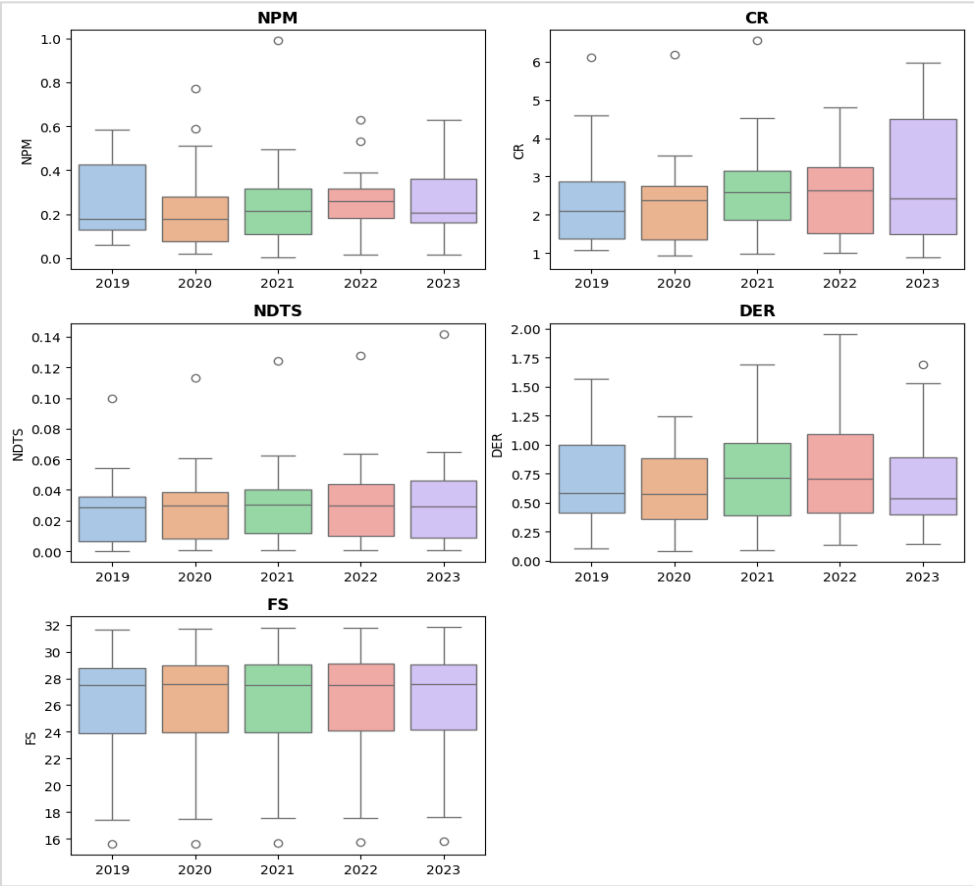


2023 period

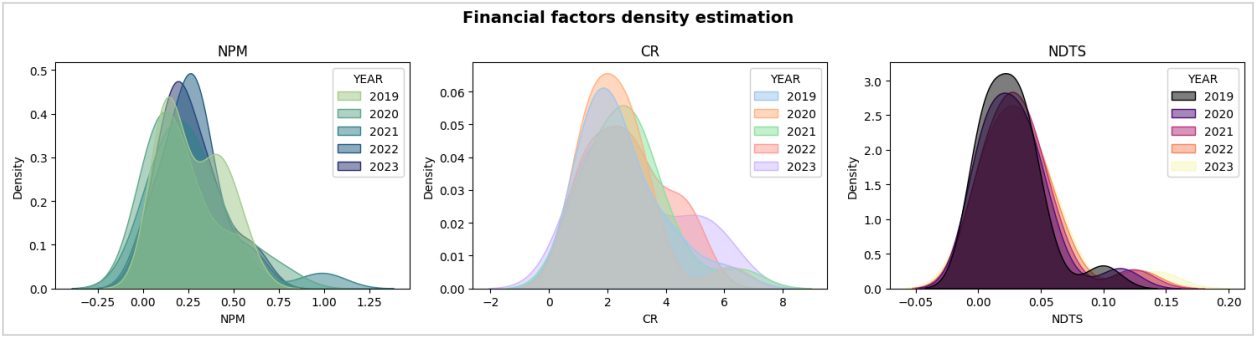
- **Moderation Factor Identification:**



- **Numerical Distribution:**



- **Independent Variable Densities:**



📈 Results & Statistical Inference

We summarize the behavior of Fixed and Random Effect models as follows:

	Fixed Effect	Random Effect
Cov. Est.	Clustered	Unadjusted
R-squared	0.0875	0.2961
R-squared (within)	0.0632	-0.2377
R-squared (between)	0.0537	0.4806
R-squared (overall)	0.0561	0.2961
F-statistic	0.9268	5.5375
p-value (F-stat)	0.4827	**0.0001**

The result above exhibits how both panel models behave. Based on the *p-value*, the Random Effects model is statistically significant, showing that the independent variables jointly explain variations in the dependent variable. In contrast, the Fixed Effects model is not statistically significant, indicating weaker explanatory power within entities.

Furthermore, the Hausman test results (*statistics*: -2.2642, *p-value*: 1.00) suggest that we fail to reject the null hypothesis, implying that the **Random Effects** estimator is appropriate for this analysis.

The Python implementation yielded the following key metrics:

Variable	Coefficient (β)	t-stat	P-value	Significance
NPM	-1.4692	-4.9722	0.0000	Significant
CR	-0.0542	-1.2745	0.2062	Insignificant
NDTS	-4.2237	-2.0110	0.0477	Significant
M_NPM	0.1329	1.3793	0.1717	Insignificant
M_CR	0.0193	1.4369	0.1547	Insignificant
M_NDTS	0.2726	0.5004	0.6182	Insignificant

Model-Level Statistics

- **Observations:** 85
- **Entities:** 17
- **Time Periods:** 5

- **Model Type:** Random Effects (PanelOLS)
- **R^2 (Overall):** 0.3435
- **R^2 (Between):** 0.5500
- **R^2 (Within):** -0.2538
- **Adjusted R-Squared**: 0.293
- **F-statistic:** 6.8893
- **Prob (F-statistic):** 0.0000 → ✅ Model is jointly significant

Hypothesis Validation

1. **Heteroskedasticity using *Breusch–Pagan***
   - **LM Stat:** 0.4164
   - **p-value:** 0.5187
   - **F-Statistic:** 0.4086
   - **F p-value:** 0.5244
   - ✅ No strong evidence of heteroskedasticity
2. **Multicollinearity via Variance Influence Factors**

Variables	VIF Score
NPM	1.840700
CR	2.166705
NDTS	2.290355
M_NPM	2.017290
M_CR	2.311458
M_NDTS	2.344806

As the table above shows that no variable is beyond threshold (VIF > 10), therefore it is examined that there is no multicollinearity issue in the model ✅.

3. **Autocorrelation by Woolridge**
   - **Lag Coefficient:** 0.1508
   - **p-value:** 0.2085
   - ✅No autocorrelation identified in this model.

4. **Normality with Shapiro-Wilk**
    - **p-value:** 0.0000
    - ❌ Not normally distributed (hypothesis is violated)
    - Normality is less critical in panel models
5. **Cross-Sectional Dependence**
    - **Average pairwise correlation:** 0.0938
    - This result indicates weak cross-sectional dependence

💡 Strategic Discussion & Conclusions

Problem Statement Explanation

1. **Do profitability, liquidity, and non-debt tax shield significantly affect capital structure individually (partially)?**

→ Based on the panel regression results with Random Effects model, variables that are consistently significant individually are profitability (NPM) and non-debt tax shield (NDTS) both in original or scaled values.

→ Liquidity (CR) does not exhibit a significant effect, and therefore cannot be considered a primary determinant in this dataset.

2. **Do profitability, liquidity, and non-debt tax shield jointly affect capital structure simultaneously?**

→ Yes. Based on the F-test, the model is jointly significant. This finding shows that the independent variables collectively influence capital structure (DER), however they do not imply that all variables are significant individually.

3. **Does the moderating variable affect the relationship either partially or simultaneously?**

→ From the estimation results, all moderator factors are not statistically significant, There is no strong evidence that firm size (FM) moderates the relationship between financial variables and capital structure.

Insight Interpretation

1. **Simultaneous vs Partial Significance**

Although the model is statistically significant altogether, only NPM and NDTS drive this result. This highlight provides insight that some independent variables could be determinators for a target variable, but cannot be a standalone factor.

2. **Liquidity is NOT the Primary Driver**

Liquidity (CR) cannot be considered a primary driver of capital structure in this model. While the negative coefficient aligns with the Pecking Order Theory, the relationship is not empirically strong.

3. **Firm Size Moderation**

Firm size does not significantly strengthen or weaken the relationship between financial variables and capital structure. Both small and large firms yield similar financing behavior.

4. **Violation in Normality Hypothesis**

The only regression hypothesis violated in this model is normality as shown by non-normally distributed residuals. Standard t-test and f-test might be less reliable, but does not invalidate the panel settings due to its meaningful descriptive interpretation.

5. **Coefficient of Determination (R-squared)**

The overall R^2 (0.3392) is much lower than commonly expected thresholds for strong explanatory power (often >0.75 in cross-sectional finance models). This tells that there are other factors beyond NPM, CR, and NDTS that contribute strongly to capital structure decisions.

Recommendations for Future Research

1. Expand the number of firms and time periods to improve statistical reliability and power.
2. Consider additional explanatory variables that capture external constraints, market conditions, and firm-specific factors.
3. Explore alternative panel methods or non-linear approaches to better capture firm behavior.